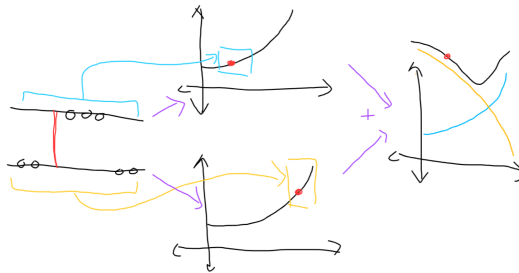


Homework 1

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Exercise 1



The inference machine works by linearly mapping the dosage to sections of the activation functions, then taking the weighted sum to get the efficacy. The dosage is first multiplied by a factor referred to as a weight, and then offset by the bias. These quantities are determined through a process called backpropagation. The result is then fed into a function referred to as an activation function; the inference machine in the video uses a softplus function. Finally, the outputs of the individual activation functions are multiplied by another factor and then added to produce the efficacy.

Exercise 2

Each atomic sentence can be either true or false. Since individual atomic sentences' truth values are independent, there are 2^n possibilities for the values of n atomic sentences, and 2^n rows are required for the truth table.